MTH2003 Tutorial Sheet 1 Solutions

1. **Question** $ty\dfrac{\mathrm{d}y}{\mathrm{d}t} = t^2 + y^2$

   Solution

   The standard form of the equation is $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \dfrac{t^2+y^2}{ty} = f(t,y)$, for $t \neq 0$ and $y \neq 0$.

   [i] This equation is not separable. To prove that, we can choose $a = 1$, $b = 1$, $c = 2$, $d = 3$, say, giving

   $$f(a,b)f(c,d) = \frac{1+1}{1} \times \frac{4+9}{6} = \frac{13}{3}$$

   while

   $$f(a,d)f(c,b) = \frac{1+9}{3} \times \frac{4+1}{2} = \frac{25}{3}$$

   whereas for separable equations we have $f(a,b)f(c,d) = f(a,d)f(c,b)$.
   [ii] This equation is homogeneous: we can rewrite it in the form

   $$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{t}{y} + \frac{y}{t} = \left(\frac{y}{t}\right)^{-1} + \frac{y}{t} = H\left(\frac{y}{t}\right).$$

   We write $y = tv$, then

   $$\frac{\mathrm{d}y}{\mathrm{d}t} = v + t\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{1}{v} + v \qquad \text{so that} \qquad t\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{1}{v}.$$

   This is a separable equation

   $$\int v\,dv = \int \frac{1}{t}\,dt$$

   $$\Rightarrow \frac{v^2}{2} = \ln(|t|) + \frac{C}{2},$$

   where C is a constant (the factor of 2 is for later convenience). Hence we have

   $$v = \pm\sqrt{2\ln(|t|) + C}$$

   $$\Rightarrow y = \pm t\sqrt{2\ln(|t|) + C}.$$

   We require that the radicand is non-negative, so that

   $$2\ln(|t|) + C \geq 0 \quad \Rightarrow \ln|t| \geq -C/2 \quad \Rightarrow |t| \geq e^{-C/2}$$

   and we note that $y = 0$ is not a solution to the ODE, so the solutions are defined on the intervals $t \in (-\infty, -e^{-C/2})$ and $t \in (e^{-C/2}, \infty)$.
   [iii] This equation is not linear, since when written in the form

   $$\frac{\mathrm{d}y}{\mathrm{d}t} - \frac{y}{t} = \frac{t}{y},$$

   we have $p(t) = -1/t$ but the right-hand side contains the unknown in the denominator.

2. (a) **Question** *Find the solution $v(x)$ of this ODE satisfying the given initial condition.*
   Solution The ODE is separable. There are no special solutions, since v on the RHS is only in the denominator. For typical solutions, we get, in the usual way,

   $$\int v\,\mathrm{d}v = -\int \frac{gR^2\,\mathrm{d}x}{(R+x)^2}, \quad x \neq -R.$$

   Integrating gives

   $$\frac{v^2}{2} = \frac{gR^2}{R+x} + c.$$

   Now we choose the integration constant so that the initial condition $v(0) = v_0$ is satisfied, i.e.

   $$\frac{v^2}{2} = \frac{gR^2}{R+x} - gR + \frac{v_0^2}{2},$$

which gives the explicit solution

$$v = \pm\sqrt{v_0^2 - 2gR + \frac{2gR^2}{R+x}}.$$

Both branches of the solutions are physically relevant: the plus sign is valid on the initial stage when the body moves upwards, and the minus sign applies later when the body falls back to Earth. For the ascending path with $v(0) = v_0 > 0$ we have

$$\boxed{v = \sqrt{v_0^2 - 2gR + \frac{2gR^2}{R+x}},}$$

and the required interval of x is where the argument of the square root is positive, which depends on the constant v_0, as discussed below. The 'minus' variant of the same formula corresponds to the later stage, when the body falls down — if it does, see below.

(b) **Question** *Analyse the interval on which this solution is defined, and the dependence of this interval on v_0. Hence determine:*

- *the expression for the minimal initial velocity $v_0 = v_*$ that is required in order that the body escapes Earth's gravitation forever. Also evaluate the numerical value to 3 s.f. for $R = 6371\,\mathrm{km}$ and $g = 9.81\,\mathrm{m/s^2}$;*
- *the velocity of the body at the infinite distance from the Earth, $v_\infty = v(x \to \infty)$, if $v_0 > v_*$;*
- *the maximal altitude x_* the body reaches if $v_0 < v_*$.*

Solution The expression under square root is positive for all $x > -R$ including $x \to \infty$, as long as $v_0^2 - 2gR \geq 0$, hence the requested minimal velocity is

$$\boxed{v_* = \sqrt{2gR}}.$$

For $R = 6371\,\mathrm{km} = 6.371 \times 10^6$ and $g = 9.81\,\mathrm{m/s^2}$, this gives $1.12 \times 10^4\,\mathrm{m/s} = 11.2\,\mathrm{km/s}$ (compare with https://en.wikipedia.org/wiki/Escape_velocity).
For $v_0 \geq v_*$, we have

$$\boxed{v_\infty = \sqrt{v_0^2 - 2gR}}.$$

For $v_0 < v_*$, the solution is defined for $x \in (-R, x_*]$ where $v(x_*) = 0$. Thus

$$v_0^2 - 2gR + \frac{2gR^2}{R+x_*} = 0,$$

wherefrom

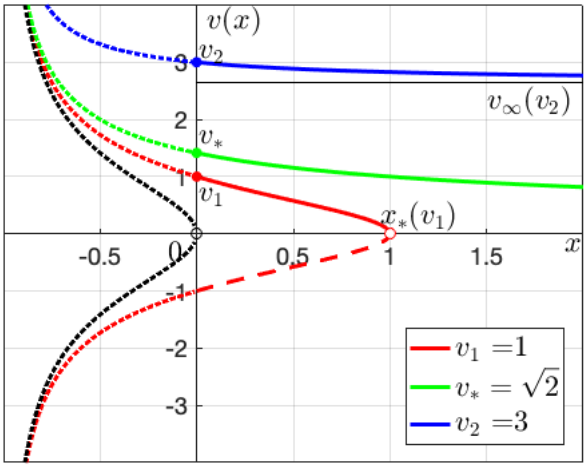
$$\boxed{x_* = \frac{2gR^2}{2gR - v_0^2} - R = \frac{Rv_0^2}{2gR - v_0^2}}$$

The above intervals for x are given for the mathematical side of the problem. Of course, in the context of the application, only $x \geq 0$ are relevant, since $x < 0$ would mean passive flight/free fall under the surface of the Earth, which should be disallowed as the assumptions of the mathematical model are valid only at or above the Earth's surface.

(c) **Question** *Sketch a representative sample of integral curves according to the following instructions:*

- *Assume that the units of measurement are chosen so that $R = 1$ and $g = 1$; use the limits $x \in [-1, 2]$, $v \in [-4, 4]$ for the plot;*
- *Plot the graphs of the solutions corresponding to $v_0 = 0$, $v_0 = v_1$, $v_0 = v_2$ and $v_0 = v_*$, where you need to choose v_1 and v_2 so that $v_1 < v_*$ and $v_2 > v_*$. You may wish to use different colours to draw graphs corresponding to different v_0.*
- *Plot also the segments corresponding to the descending part of the body path.*
- *On the coordinate axes, indicate the special points corresponding to the selected solutions, i.e. v_0 and $v_\infty(v_0)$ on the ordinate axis and $x_*(v_0)$ on the abscissa axis, where appropriate.*

Solution For the given parameters $g = 1$, $R = 1$, we have $v_* = \sqrt{2}$. The graph below is for the choice $v_1 = 1 < v_*$ and $v_2 = 3 > v_*$. The choice of colours: black for $v_0 = 0$; red for $v_0 = v_1$, green for $v_0 = v_*$ and blue for $v_0 = v_2$. The pieces of graphs for $x > 0$, $v > 0$ (ascending) are solid lines, while $v < 0$ (descending) are dashed lines. The physically unrealistic $x < 0$ (before the start, "underground") are dotted lines.


$v(x)$
v_2
$v_\infty(v_2)$
v_*
v_1
$x_*(v_1)$
x
3
2
1
0
-1
-2
-3
-0.5
0
0.5
1
1.5
$v_1 = 1$
$v_* = \sqrt{2}$
$v_2 = 3$